

Math 242, S09
Quiz 3

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (5 points) Find a general solution for the first-order differential equation

$$\frac{dy}{dx} = 2x \sec(y).$$

Solution: $\frac{1}{\sec y} dy = 2x dx$
 $\int \cos y dy = \int 2x dx$
 $\sin y = x^2 + C$
 $y = \arcsin(x^2 + C)$

Problem 2. (5 points) Find the particular solution of the initial value problem:

$$2y \frac{dy}{dx} = \frac{x}{\sqrt{x^2 - 16}}, \quad y(5) = 2.$$

Solution: $\int 2y dy = \int \frac{x}{\sqrt{x^2 - 16}} dx$
 $y^2 = \sqrt{x^2 - 16} + C \quad \leftarrow \text{General solution}$
 $y(5) = 2 \Rightarrow 2^2 = \sqrt{5^2 - 16} + C$
 $\Rightarrow C = 4 - 3 = 1$
 $\Rightarrow \text{the particular solution is given by}$
 $y^2 - \sqrt{x^2 - 16} - 1 = 0$